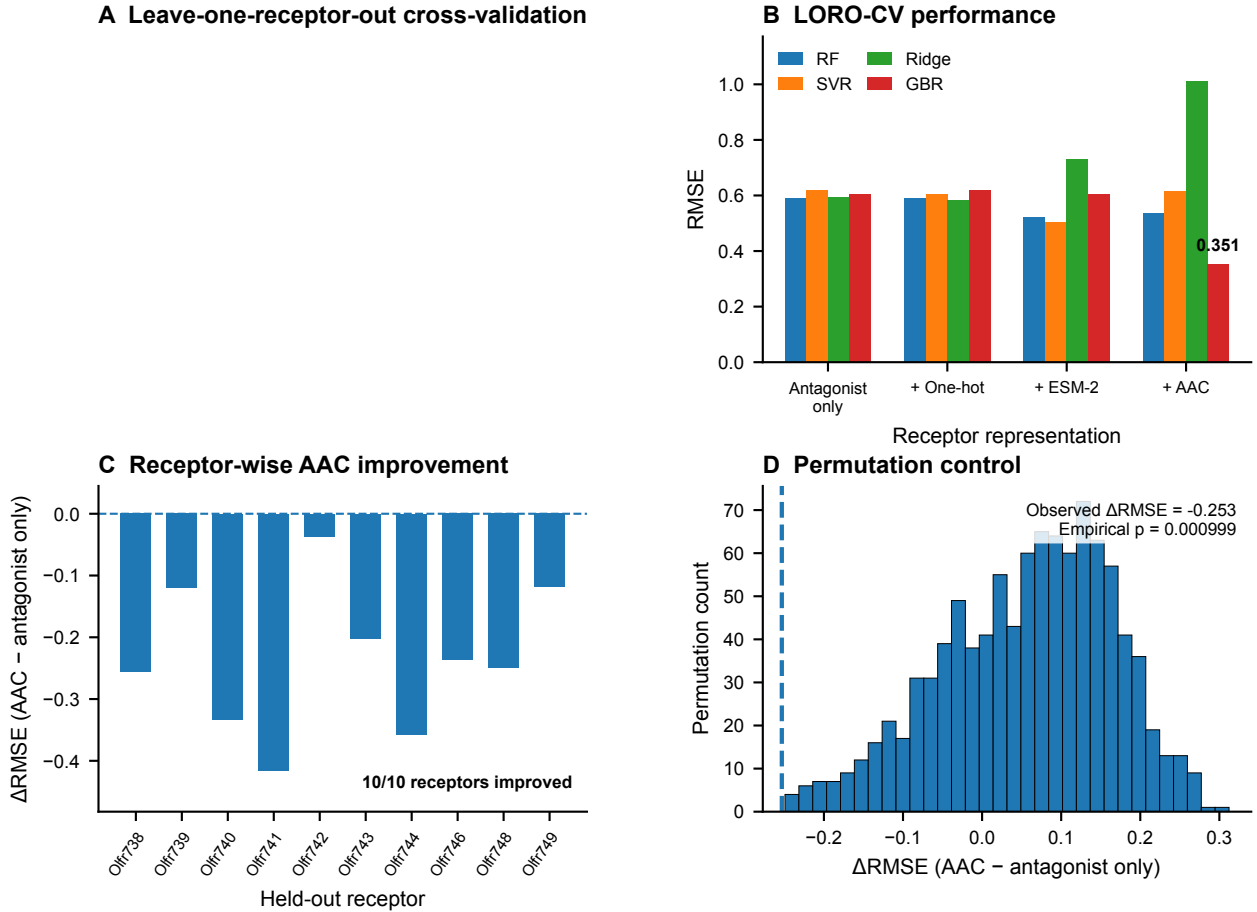
A Leave-one-receptor-out cross-validation
B LORO-CV performance
RF
SVR
Ridge
GBR
RMSE
1.0
0.8
0.6
0.4
0.2
0.0
0.351
Antagonist only
+ One-hot
+ ESM-2
+ AAC
Receptor representation
C Receptor-wise AAC improvement
ΔRMSE (AAC − antagonist only)
0.0
−0.1
−0.2
−0.3
−0.4
10/10 receptors improved
Olfr738
Olfr739
Olfr740
Olfr741
Olfr742
Olfr743
Olfr744
Olfr746
Olfr748
Olfr749
Held-out receptor
D Permutation control
Observed ΔRMSE = -0.253
Empirical p = 0.000999
Permutation count
70
60
50
40
30
20
10
0
−0.2
−0.1
0.0
0.1
0.2
0.3
ΔRMSE (AAC − antagonist only)